

CPRReady: Proposed Application Behavior and Session Management Protocol

Purpose: This document outlines the proposed application behavior for Bluetooth disconnection, navigation restrictions, session results and History, post-session navigation, and pre-session neutral-position verification.

The proposed protocols are intended to simplify the initial CPRReady prototype while maintaining the reliability and integrity of the recorded CPR measurements. Development will prioritize one complete, instructor-supervised practice flow. Firmware-dependent functions remain subject to technical consultation, implementation and validation.

1. Bluetooth Disconnection During an Active Session

Proposed Protocol

If the Bluetooth Low Energy (BLE) connection is lost during an active practice session, Flutter will terminate the session on the application side and classify it as incomplete. The instructor must stop the practice, reconnect the device and initiate a new session.

Automatic reconnection and continuation of an interrupted session will not be prioritized for the initial prototype.

Previously received measurements will not be displayed or recorded as current or finalized results after a connection loss.

Proposed Application Flow

1. Flutter detects that the BLE connection has been lost during Live Practice.
2. The application immediately stops real-time visual and voice feedback and clears the displayed CPR measurements to prevent stale data from being presented.
3. Flutter classifies the active session as incomplete and opens the **Session Interrupted** screen.
4. The screen displays the notification: **“Connection lost. Practice session interrupted.”** No CPR measurements or finalized results are displayed.

5. The instructor stops the physical practice and selects one of the following navigation options:
 - **Reconnect Device:** Opens the Bluetooth connection process. Once the connection is reestablished, the instructor may initiate a new practice session.
 - **Return to Home:** Returns to the Home screen.
6. The interrupted session is not saved to the completed-session History.

The application will not attempt to resume the interrupted session after reconnection.

Rationale and Technical Considerations

Allowing an interrupted session to resume introduces additional requirements for data preservation and synchronization between Flutter and the ESP32.

Although the ESP32 may continue collecting measurements during a Bluetooth interruption, Flutter will no longer receive the transmitted data. Unless the firmware can retain the complete session and recover any missing measurements, reconnection may result in gaps or inconsistencies in the application's records.

These interruptions could affect the completeness of the recorded measurements, particularly when calculating session-level indicators such as chest compression fraction (CCF).

Terminating the interrupted session provides a simpler approach for the initial prototype and avoids treating potentially incomplete measurements as valid results.

However, terminating the session on the application side does not guarantee that the ESP32 has stopped measuring. Flutter cannot reliably transmit a **STOP** command while the Bluetooth connection is lost.

For Technical Verification: The firmware should ideally detect a BLE disconnection and terminate its active session. The hardware team must confirm how the ESP32 will handle disconnection and whether its session state will reset before another practice session begins.

2. Navigation Restrictions During Live Practice

Proposed Protocol

For the initial prototype, CPReady is intended to operate in the foreground during active practice sessions.

Navigation to other application pages will be restricted while a practice session is active. This includes the Home, Practice and History tabs, as well as the phone's Back action.

The instructor must use the designated **End Session** control to complete the practice normally. Navigation will become available again once the session has been completed or classified as interrupted.

Continuous background operation will not be a required feature for the initial prototype.

Proposed Application Flow

1. The instructor begins a practice session.
2. Flutter opens the Live Practice interface and restricts navigation to other application pages.
3. If the instructor attempts to navigate to another page within CPReady or use the Back action, the application prevents the transition while maintaining the active session.
4. The instructor may complete the session using the designated press-and-hold End Session button.
5. Once the session is completed or classified as interrupted, the navigation restrictions are removed.
6. The instructor may then use the navigation options provided on the corresponding Results or interruption screen.

Rationale and Technical Considerations

Restricting navigation reduces the possibility of accidentally leaving the practice interface while the ESP32 is collecting and transmitting measurements.

Allowing unrestricted navigation during an active session would require additional handling for session-state preservation, continued BLE communication and the behavior of the application across different screens.

These requirements would increase development complexity without directly contributing to the essential functions of the initial prototype.

Keeping the instructor on the Live Practice interface provides a more manageable application flow and reduces the risk of accidental interruptions.

Limitation: Application Minimization and Unexpected Termination

Flutter can restrict navigation within CPReady but cannot completely prevent the phone's operating system from minimizing or terminating the application.

Continuous background operation will not be a required feature for the initial prototype. If the application is minimized, session continuity and uninterrupted BLE communication cannot be guaranteed.

Upon returning, Flutter may continue the session only if its active state and BLE connection remain intact and fresh measurements are available. Otherwise, the session will be classified as interrupted, and previously received measurements will not be displayed or recorded as current results.

If the application is terminated unexpectedly, the unfinished session must not be treated as successfully completed.

If the application can detect the interrupted session upon reopening, it may display an appropriate interruption notice. Otherwise, the application may return to Home without creating a completed-session record.

For Technical Verification: The application team must test how Flutter behaves when CPReady is minimized and reopened. The hardware team must also confirm how the ESP32 will handle the resulting communication interruption and reset its session state before another practice begins.

3. Session Results, Automatic Saving and History

Proposed Protocol

Only successfully completed sessions with valid final results will be automatically saved to the application's local History.

The ESP32 will be responsible for calculating and finalizing the CPR performance indicators, including compression depth, compression rate, chest recoil and chest compression fraction (CCF). Flutter will receive these finalized measurements, validate the transmitted data and automatically save the completed session.

The instructor will not be required to manually confirm whether the results should be saved. Once the finalized results have been successfully received, validated and saved, the application will automatically open the Results screen.

Sessions interrupted by Bluetooth disconnection and sessions with missing or invalid final results will not be saved to History. These situations will display their respective error screens without presenting previously received measurements as current or finalized results.

History will contain a single chronological list of successfully completed and saved practice sessions. Interruptions and system failures will be documented separately during research testing to support the evaluation of system functionality.

Proposed Application Flow

A. Successfully Completed Session

1. The instructor completes the press-and-hold End Session interaction while the trainee continues performing compressions.
2. Flutter transmits the **STOP** command to the ESP32.
3. Upon accepting the command, the ESP32 closes the measurement period and finalizes the session.
4. The ESP32 transmits the finalized depth, rate, recoil and CCF results to Flutter.
5. Flutter receives and validates the complete set of finalized measurements.
6. The application automatically saves the completed session to local storage.
7. Once saving succeeds, Flutter opens the Results screen and displays the finalized measurements.
8. The instructor reviews the results and selects one of the following navigation options:
 - **Start New Practice:** Returns to Practice Setup to begin another session. The existing Bluetooth connection may be reused if it remains active.
 - **Return to Home:** Returns to the Home screen, where the instructor can access Practice or History.
9. The completed session remains accessible through History regardless of the selected navigation option.

B. Session Interrupted by Bluetooth Disconnection

1. Flutter detects that the Bluetooth connection has been lost during an active practice session.
2. The application immediately stops real-time visual and voice feedback and clears the displayed measurements to prevent stale data from being presented.
3. The session is classified as incomplete, and Flutter opens the **Session Interrupted** screen.
4. The screen notifies the instructor that the Bluetooth connection was lost and that the practice session could not be completed. No CPR measurements or finalized results are displayed.
5. The instructor selects one of the following navigation options:
 - **Reconnect Device:** Opens the Bluetooth connection process. Once the connection is reestablished, the instructor may initiate a new practice session.
 - **Return to Home:** Returns to the Home screen.

6. The interrupted session is not saved to History.

The application will not attempt to resume the interrupted session. The ESP32's response to Bluetooth disconnection, including whether it stops and resets its active session, must be confirmed by the hardware team.

C. Missing or Invalid Final Results

1. The session reaches the ending process, but Flutter does not receive a complete and valid set of finalized results from the ESP32.
2. If the expected final results are not received within an agreed response period, or if the received data fail validation, the application opens the **Results Unavailable** screen.
3. The screen notifies the instructor that the final results could not be retrieved or processed. No CPR measurements or finalized results are displayed.
4. The instructor selects **Return to Home**, where another practice session may be initiated.
5. The session is not classified as successfully completed and is not saved to History.

Missing or invalid results refer to technical problems with the final data rather than poor CPR performance. Examples include a final-results packet that is not received, an incomplete set of required measurements, a packet that cannot be decoded using the agreed data format, or a measurement containing an impossible value.

Measurements that indicate poor CPR performance, such as insufficient compression depth or an excessively fast compression rate, are still valid results if they have been correctly calculated and transmitted.

Although the ESP32 may have successfully finalized the session, Flutter cannot display or save results that it has not received in a complete and usable form.

A result-retrieval function may be considered if the firmware can retain and retransmit finalized measurements. However, this capability will not be assumed for the initial prototype.

For Technical Verification: The hardware and application teams must establish an appropriate response timeout and agree on how Flutter will identify missing, incomplete or invalid final-results transmissions.

D. Local Saving Failure

1. Flutter successfully receives and validates the complete set of finalized results but encounters an error while saving the record to local storage.
2. The application temporarily retains the valid results in memory and opens the **Saving Error** screen.
3. The screen notifies the instructor that the results have been received but have not been successfully saved.
4. The instructor selects **Retry Saving** to attempt the local storage operation again.

5. Once saving succeeds, Flutter opens the Results screen and adds the completed session to History.

The application must not indicate that the results have been saved until the local storage operation has succeeded.

If saving repeatedly fails, the instructor must be informed that the record remains unsaved. Temporarily retained results may be lost if the application is terminated before saving is completed.

The application team must determine the final behavior for leaving the Saving Error screen during implementation.

Session Storage and Navigation Rules

Session outcome	Screen displayed	Navigation	Save to History
Completed session with valid final results and successful saving	Results	Start New Practice or Return to Home	Yes
Bluetooth disconnection during practice	Session Interrupted	Reconnect Device or Return to Home	No
Unexpected application termination during practice	Interruption notice upon reopening, if detectable	Return to Home and begin a new session	No
Missing or invalid final results	Results Unavailable	Return to Home	No

Valid final results received but local saving fails	Saving Error	Retry Saving	Only after successful saving
---	--------------	--------------	------------------------------

The Results screen will be displayed only when the session has been successfully finalized, its complete results have been received and validated, and the record has been saved.

The Session Interrupted, Results Unavailable and Saving Error screens serve as notifications and recovery interfaces. They will not display stale CPR measurements or be treated as completed-session records.

Rationale and Technical Considerations

Automatic saving reduces the number of actions required from the instructor and ensures that successfully completed practice sessions are recorded without an additional saving confirmation.

Maintaining a single History page simplifies the application interface and allows the instructor to access completed sessions through a chronological list. Selecting a record will open its Session Details page, where the saved CPR performance indicators can be reviewed.

Excluding interrupted sessions and sessions with missing or invalid final results prevents incomplete or unreliable measurements from being presented as successfully completed practice records. The primary consideration is data integrity rather than storage capacity.

The application must also distinguish between **invalid data** and **poor CPR performance**. Measurements that fall outside the recommended CPR performance ranges must still be recorded when they are valid outputs of a successfully completed session. Such measurements are essential for evaluating the trainee's performance.

Previously received measurements must not be displayed or recorded as current or finalized results following a Bluetooth disconnection or a failure to retrieve the final results. This prevents the application from presenting stale data that may not represent the completed measurement period.

Although incomplete sessions will be excluded from the application's History, they must still be documented during research testing. Bluetooth interruptions, missing results, failed transmissions and data-logging errors should be recorded separately in the research data collection sheet.

This distinction allows History to function as a record of completed practice sessions while ensuring that unsuccessful operations are still considered when evaluating the system's functionality and reliability.

For Technical Verification: The hardware and application teams must confirm the following requirements:

1. The ESP32 must provide a reliable indication that it has accepted the **STOP** command, closed the measurement period and finalized the session.
2. The ESP32 must transmit a complete set of finalized depth, rate, recoil and CCF results using an agreed data format.
3. Flutter must be able to distinguish finalized results from ordinary real-time measurement updates.
4. The teams must establish how missing, incomplete or malformed final-results transmissions will be identified.
5. The firmware must define its response to Bluetooth disconnection and unexpected application termination, particularly how an active session will be stopped or reset.
6. Any proposed capability to retain and retransmit finalized results must be confirmed before a result-retrieval function is included in the application.

4. Post-Session Navigation

Proposed Protocol

After a completed session has been successfully saved, Flutter will automatically display the Results screen.

The Results screen will remain open until the instructor selects a navigation option. A separate Save button or saving confirmation will not be required.

Two navigation options will be provided: **Start New Practice** and **Return to Home**.

Proposed Application Flow

1. Flutter successfully saves the finalized session results.
2. The application opens the Results screen and displays the trainee's recorded depth, rate, recoil and CCF results.
3. The instructor reviews the results and selects one of the following options:
 - **Start New Practice:** Returns to Practice Setup, where the instructor can enter the next trainee's information and complete the required preparation checks. The existing Bluetooth connection may be reused if it remains active.

- **Return to Home:** Returns to the Home screen, where the instructor can access Practice or History.
- 4. The completed session remains accessible through History regardless of the selected navigation option.

Rationale and Technical Considerations

Automatic saving removes an unnecessary action from the post-session process while ensuring that completed practice records are retained.

Providing navigation options allows the instructor to begin another practice session or return to the main application interface after reviewing the results.

A separate View History button is not required on the Results screen because the completed session has already been saved and History remains accessible through the application's navigation.

For the initial prototype, limiting the Results screen to two primary navigation options provides a straightforward post-session experience.

5. Pre-Session Neutral-Position Verification

Proposed Protocol

For the initial prototype, the trainee's neutral position will be verified by the instructor before the ESP32 performs its baseline calibration.

Automated detection of pre-session leaning will not be a mandatory feature unless the hardware team can establish and validate a reliable detection method within the available development period.

Proposed Application Flow

1. The trainee completes the required preparation steps and assumes the correct CPR position.
2. The instructor verifies the trainee's hand placement.
3. The instructor checks that the trainee is not visibly compressing the manikin's chest.
4. The trainee maintains the correct hand placement without applying downward pressure.
5. The instructor confirms readiness and selects **Begin Session**.

6. The ESP32 performs its baseline calibration while the trainee maintains the neutral position.
7. Once calibration is successfully completed, the application provides the signal to begin compressions.
8. The trainee starts performing chest compressions, and the application begins presenting real-time feedback.

Rationale and Technical Considerations

The proposed hardware configuration uses two MPU6050 sensors: one positioned on top of the manikin and the other beneath it.

Although the sensors can provide acceleration and orientation-related measurements, they do not directly measure the static force applied to the manikin's chest.

Consequently, automated detection of pre-existing leaning would require a validated method for establishing a neutral reference position and distinguishing actual chest compression from ordinary movement, sensor noise and minor changes in hand position.

Implementing automated leaning detection without adequate validation could introduce false warnings or allow an incorrect baseline to be accepted.

Instructor verification offers a simpler preparation procedure for the initial prototype. However, it also has limitations.

An instructor may be able to identify visibly incorrect hand placement or obvious downward pressure, but visual observation cannot establish an exact neutral position or reliably detect every instance of residual pressure.

Therefore, instructor verification should be treated as a **standardized preparation procedure rather than an objective measurement of the neutral position**.

The ESP32's subsequent calibration remains necessary, and its ability to establish a dependable baseline must be evaluated during hardware validation.

For Technical Verification: The hardware team must confirm the calibration procedure, the conditions required for successful calibration and how the ESP32 will communicate calibration completion or failure to Flutter.

If a reliable automated neutral-position check becomes feasible, it may be incorporated into the application flow. Otherwise, the initial prototype will retain instructor verification, with its limitations acknowledged during evaluation.

6. Summary of Proposed Decisions

Area	Proposed decision
Bluetooth disconnection	Terminate the app's active session, classify it as incomplete and require a new session after reconnection.
Navigation during practice	Restrict navigation within CPReady during Live Practice and restore it after completion or interruption.
Background operation	Not required for the initial prototype. Continue after minimization only if session continuity, BLE connectivity and fresh measurements are confirmed.
Unexpected application termination	Do not classify the unfinished session as successfully completed.
Completed-session results	Automatically save valid final results and open the Results screen.
Post-session navigation	Provide Start New Practice and Return to Home options.
Incomplete sessions	Exclude from completed-session History but document interruptions separately for research evaluation.

Missing or invalid results	Display Results Unavailable without stale measurements and do not classify the session as successfully completed.
Local saving failure	Temporarily retain valid results and allow the instructor to retry saving.
Neutral-position verification	Use instructor verification followed by ESP32 baseline calibration for the initial prototype.
Automated leaning detection	Include only if the hardware team can implement and validate a reliable method.